LUX PROVIDE

SCynergy 2026

Getting Started with MeluXina

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HPC Software Engineer

EPICURE
Unlocking European-level HPC Support

MADE IN LUXEMBOURG

Getting Started with MeluXina

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High Performance Computing

- What is HPC?

HPC Platforms

- Hardware view
- Software view

Using an HPC platform

- User Interfaces
- Interactive vs batch job
- Job scheduler
- Software environment

MeluXina in practice

- User and project accounts
- Data storage
- Authentication
- Shell & web-portal access

Hands-on!

- Configuration
- Urban wind simulation
- Deep learning with PyTorch

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What is

High-Performance Computing?

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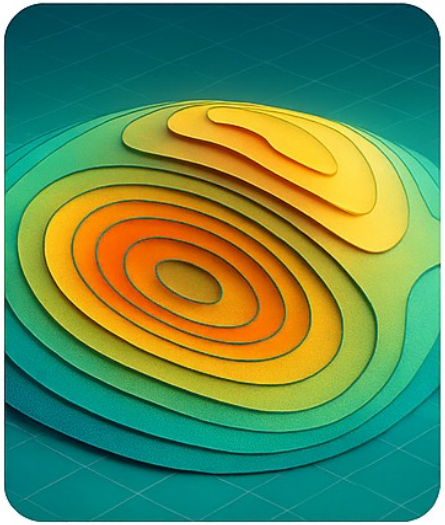
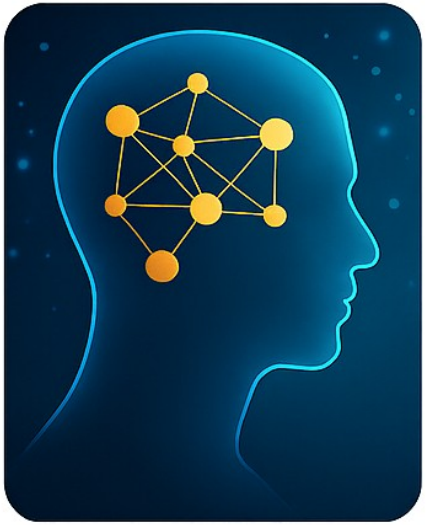
HPC is everywhere



Simulations & modeling



AI / ML training & inference

Large-scale data analytics

Engineering, science & industrial R & D

What is High Performance Computing?

High Performance Computing (HPC)

- Use of **parallel and distributed computers with fast interconnects**
- To execute an application quickly and efficiently

Why parallel computers?

- Performance of single CPU core is getting limited (power, physics)
- Multiple cores are used to increase the computing capacity

HPC is challenging

- Active research domain
- Provides tools for industry and research

How to get faster with HPC?

~~Build faster processors~~

- Moore's law continues but ***The free lunch is over!***
- CPU serial-processing speed is reaching its physical limit

➔ **Multi-cores processor architectures**

➔ **Accelerators and specialized processors** (GPU, TPU, FPGA, etc.)

Combine multiple computers

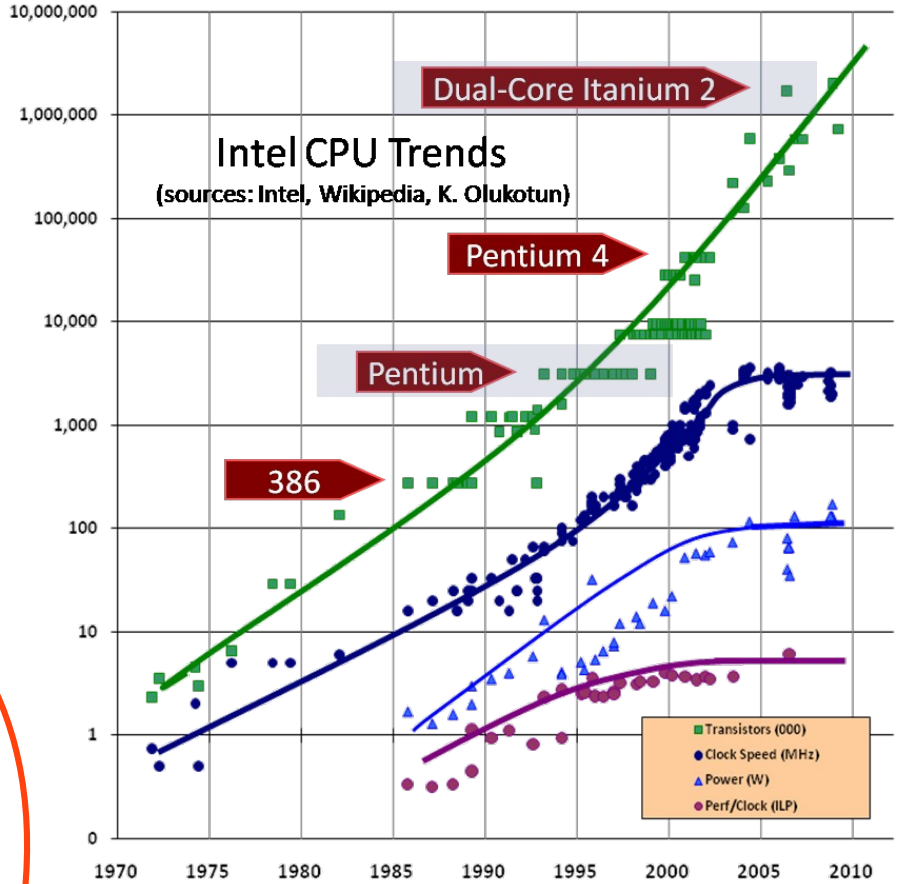
➔ **HPC Clusters and Supercomputers**

Better use of the hardware

➔ **Identify the actual bottleneck** (CPU, memory, network, etc.)

➔ **Vectorization (SIMD)**

Not to forget: **Better algorithms**



H. Sutter. *The Free Lunch Is Over*. 2005

Parallel programming

How much faster is HPC?

	Your laptop	MeluXina (shared with other users)	EL CAPITAN (shared with other users)
CPU	8 cores	89,984 cores	1,051,392 cores
Memory	32 **G**B	488 **T**B	5.44 **P**B
Storage	1 **T**B	12.6 **P**B	TBA
Network	Ethernet 10 Gb/s	Infiniband 200 Gb/s	Slingshot 200 Gb/s
Accelerators	1 GPU	800 GPUs	43,808 GPUs
Theoretical Performance	350 **G**flops	18 **P**flops	2.82 **E**flops

HPC Platforms

Hardware and Software Views

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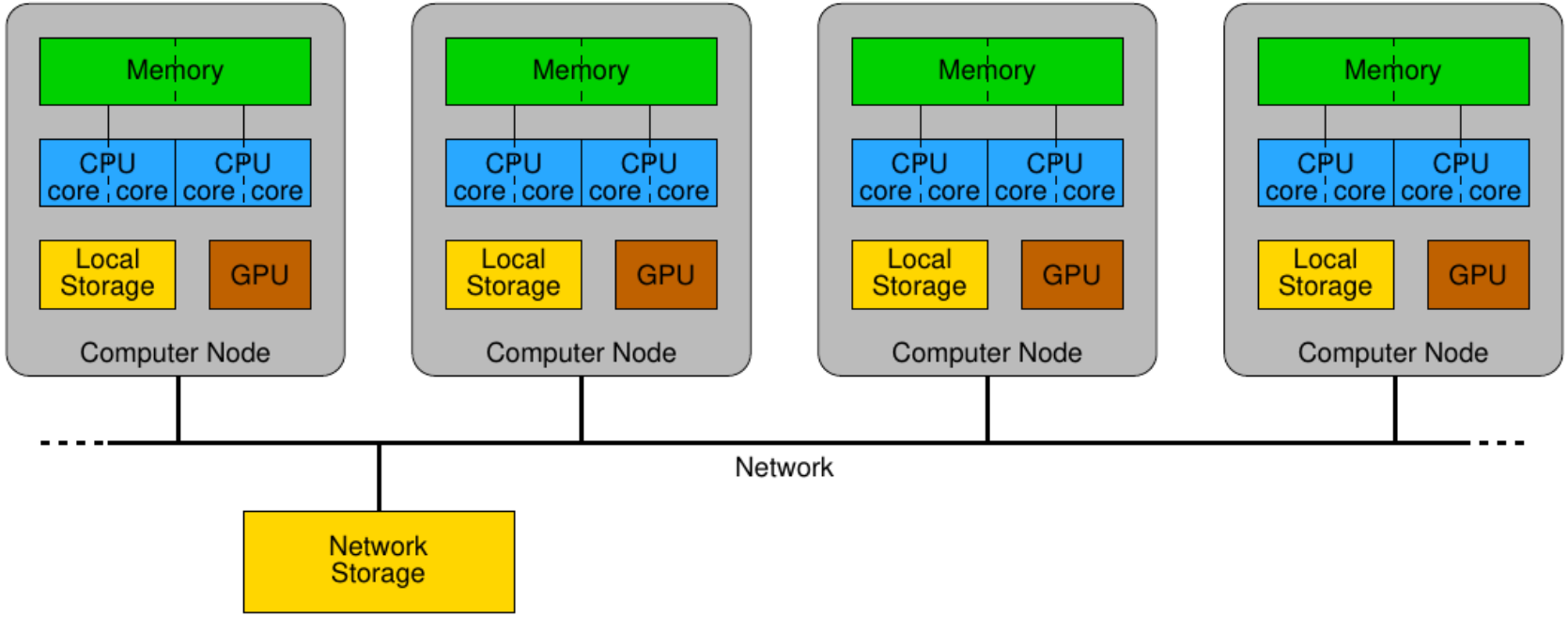
What does an HPC platform look like?

Photos: Garry McLeod/LLNL

El Capitan, Lawrence Livermore National Laboratory, USA

MeluXina, LuxProvide,

HPC platform: Hardware View



Different components or type of **resources**:

- Processors, GPU/Accelerators, Memory, Storage, Network, ...

HPC platform: Software View

Operating System

- Manage hardware resources
- Interface between userspace and hardware

HPC Tools

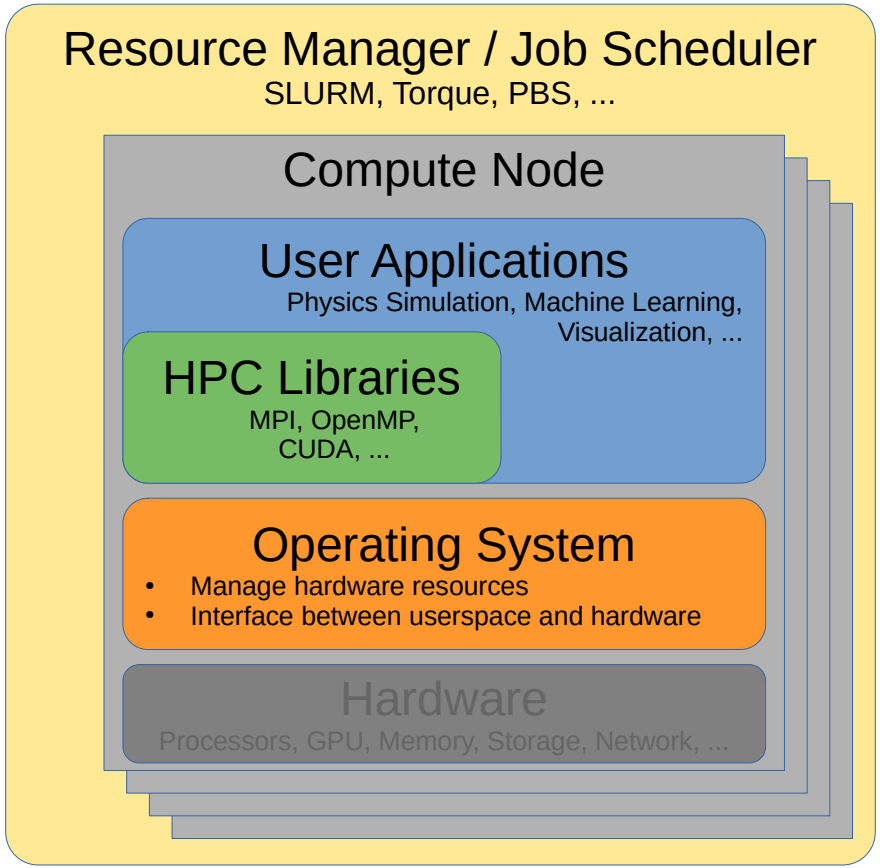
- Libraries, compilers and tools for developing and running parallel programs
- MPI, OpenMP, CUDA, TensorFlow, ...

User Applications

- Scientific, engineering, and commercial
- Simulation, Machine Learning, Visualization, ...

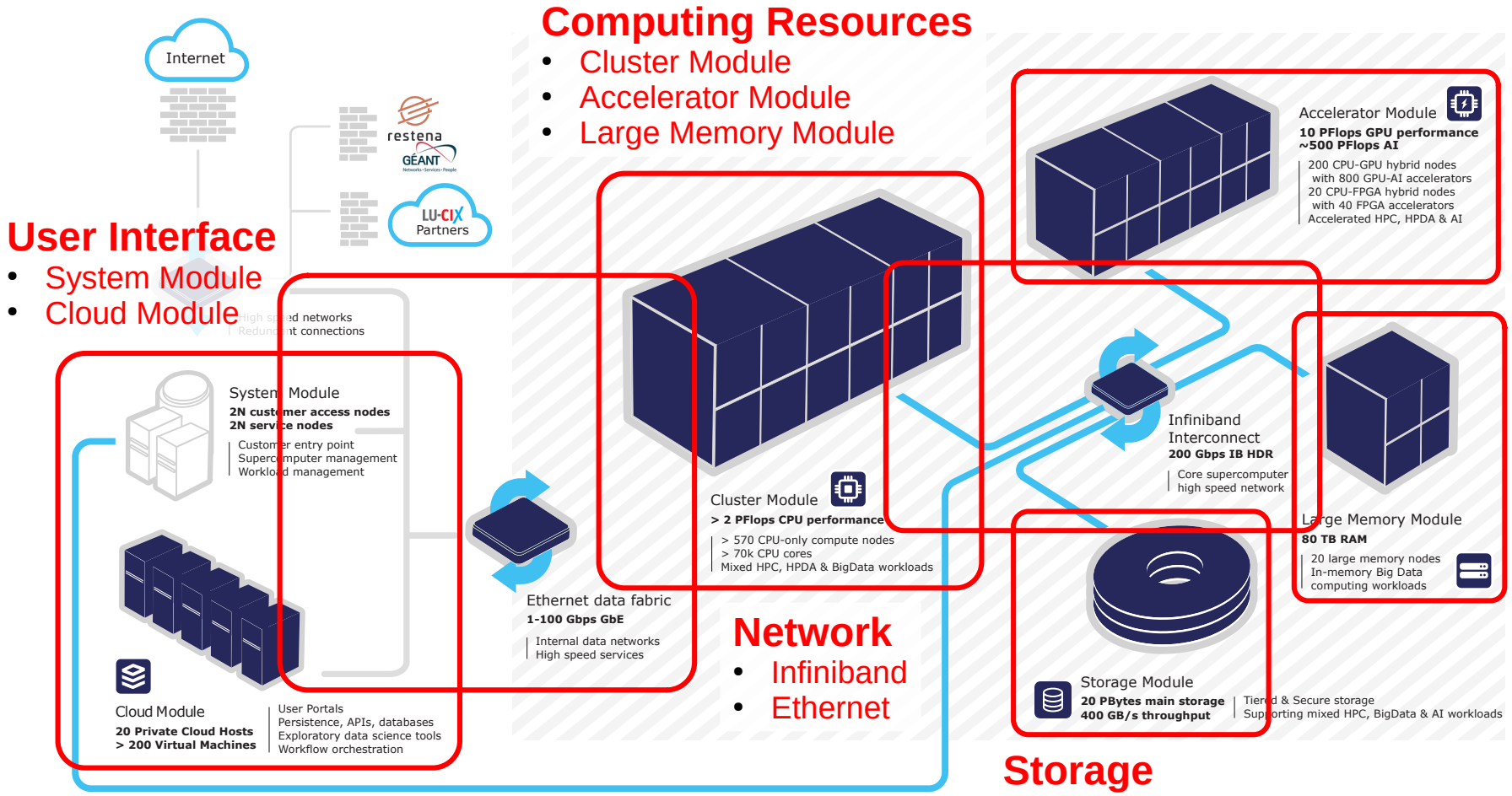
Resource Manager / Job Scheduler

- Allocate resources and schedule user jobs
- Ensure efficient and fair resource sharing



Example: MeluXina Supercomputer

Source: https://docs.lxp.lu/system/overview/



Using an HPC Platform

HPC from the user point of view

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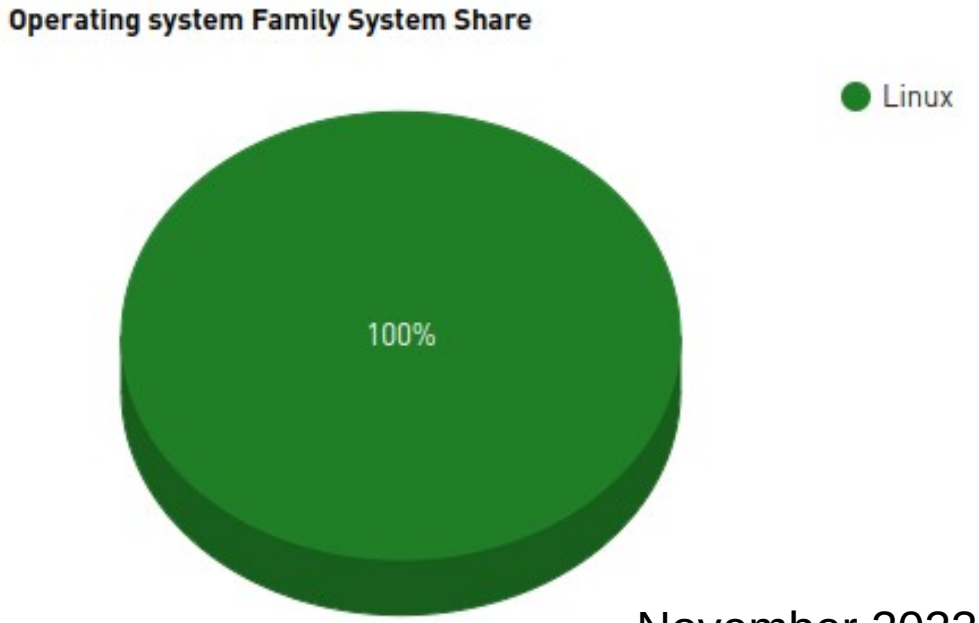
Quiz!

What is the Operating System used on the Top500 supercomputers?



November 2023

Source: https://www.top500.org/statistics/list/

User Interfaces for HPC

Command Line Interface (CLI)

- By far the most common

Advantages

- Simple (always works)
- Easy to program/automatize
- Fast and low bandwidth requirements

Disadvantages

- Not user friendly
- Requires knowledge and memorizing

Alternatives to CLI

Web portal

- Dedicated to some specific tasks

Remote connection

- Client GUI and remote server for dedicated applications

API for remote execution

GUI for dedicated tasks

- For example, file transfer

Command Line Interface

xbesseron@access2:~/202401-Large_Midrex_Blast_Furnace_Testing/run_te...0_2024-02-02_17-39-11_LargeMidrexOF_OF128-simple-dt0.00001 — Konsole

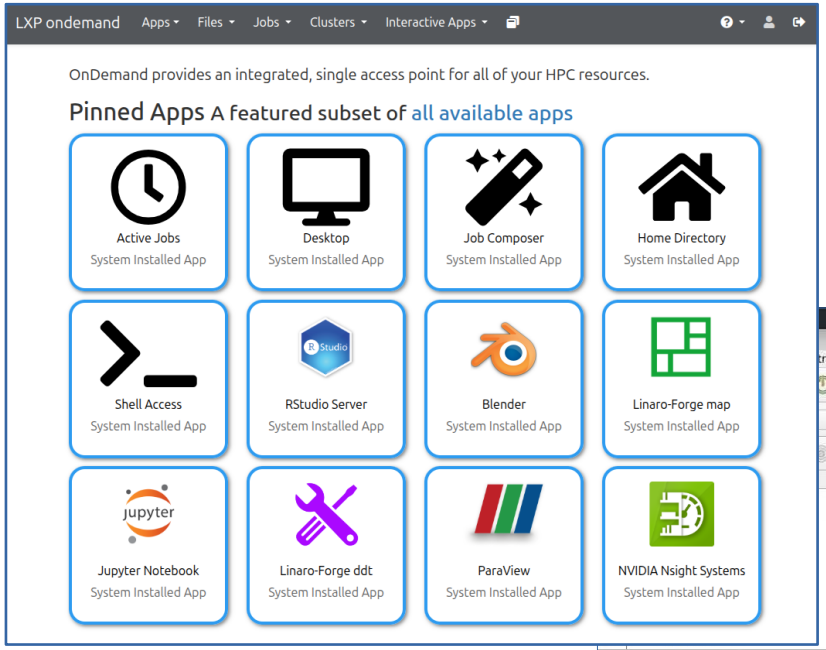
```
DILUPBiCGStab:  Solving for CH4, Initial residual = 4.1610352e-07, Final residual = 4.1610352e-07, No Iterations 0
DILUPBiCGStab:  Solving for CO, Initial residual = 4.1610352e-07, Final residual = 4.1610352e-07, No Iterations 0
DILUPBiCGStab:  Solving for CO2, Initial residual = 4.1610352e-07, Final residual = 4.1610352e-07, No Iterations 0
DILUPBiCGStab:  Solving for H2, Initial residual = 4.1610352e-07, Final residual = 4.1610352e-07, No Iterations 0
DILUPBiCGStab:  Solving for H2O, Initial residual = 4.1610352e-07, Final residual = 4.1610352e-07, No Iterations 0
DILUPBiCGStab:  Solving for O2, Initial residual = 4.1610352e-07, Final residual = 4.1610352e-07, No Iterations 0
DILUPBiCGStab:  Solving for h, Initial residual = 0.00019233561, Final residual = 3.0765688e-09, No Iterations 1
min/max(T) = 1161.0753, 1175.5424
DICPCG:  Solving for p, Initial residual = 8.4633275e-05, Final residual = 7.2521317e-07, No Iterations 1
diagonal:  Solving for rho, Initial residual = 0, Final residual = 0, No Iterations 0
time step continuity errors : sum local = 1.5739e-11, global = 2.2920826e-13, cumulative = 3.8700406e-09
DICPCG:  Solving for p, Initial residual = 7.2976685e-07, Final residual = 7.2976685e-07, No Iterations 0
diagonal:  Solving for rho, Initial residual = 0, Final residual = 0, No Iterations 0
time step continuity errors : sum local = 1.5837742e-11, global = 2.1096539e-13, cumulative = 3.8702516e-09
DICPCG:  Solving for p, Initial residual = 7.2983852e-07, Final residual = 7.2983852e-07, No Iterations 0
diagonal:  Solving for rho, Initial residual = 0, Final residual = 0, No Iterations 0
time step continuity errors : sum local = 1.5839298e-11, global = 2.2920826e-13, cumulative = 3.8704808e-09
ExecutionTime = 113.93 s  ClockT

Courant Number mean: 5.3197731e-
Time = 0.00054

diagonal:  Solving for rho, Init
PIMPLE: Iteration 1
```

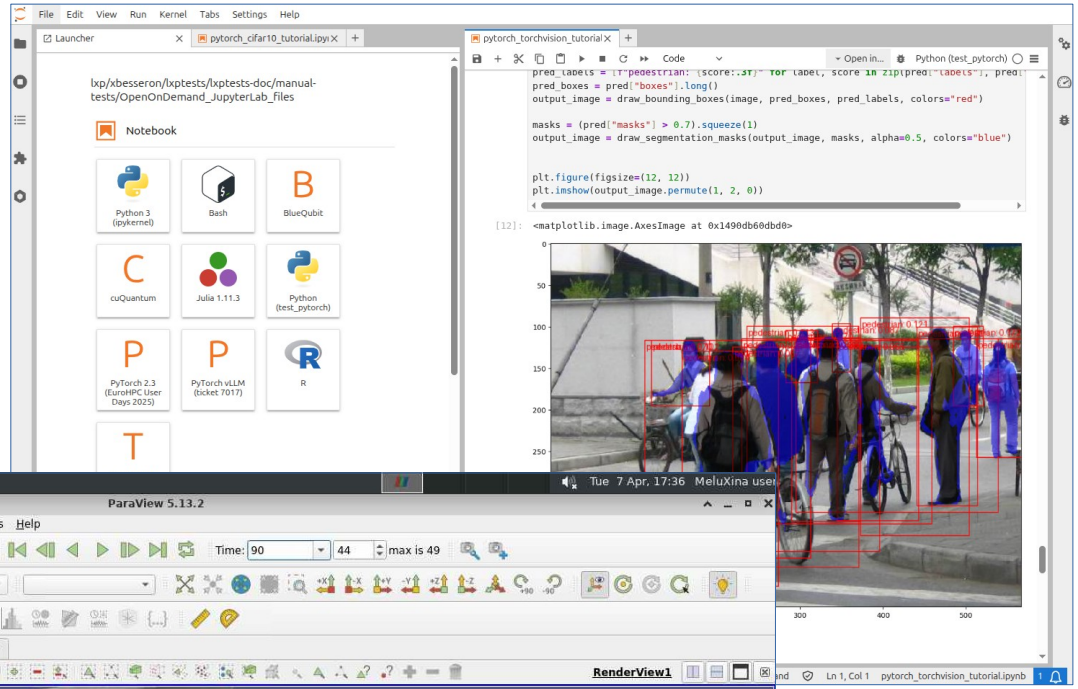
xbesseron@aion-0229:~ — Konsole

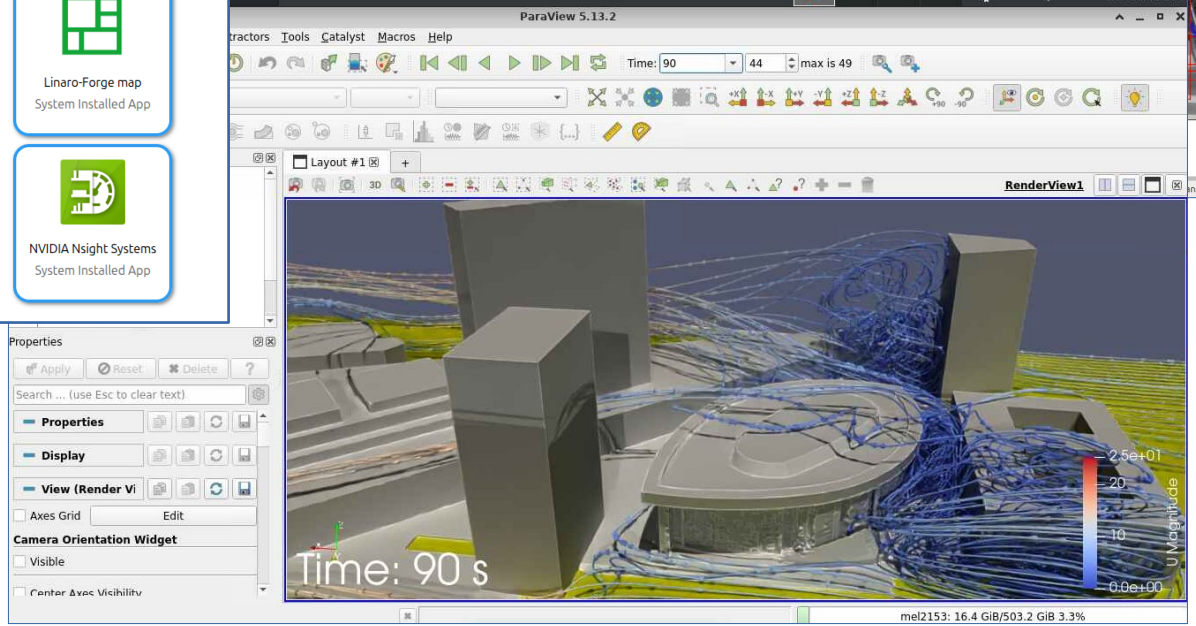
u100054@login02:~ — Konsole

```
MeluXina

.-----------------------------------------------------------------------.
| You are on a MeluXina login node                                      |
|-----------------------------------------------------------------------|
|                      System information: Compute                      |
|-----------------------------------------------------------------------|
| Nodes | CPU                      | RAM    | Accelerator              | Disk  |
|-------|--------------------------|--------|--------------------------|-------|
| 573N  | 2x AMD 7H12: 128c @2.6G  |  512GB | -                        | -     |
| 200N  | 2x AMD 7452:  64c @2.3G  |  512GB | 4x NVIDIA A100-40        | 1.92G |
|  20N  | 2x AMD 7452:  64c @2.3G  |  512GB | 2x Intel Stratix 10MX-16 | 1.92G |
|  20N  | 2x AMD 7H12: 128c @2.6G  | 4096GB | -                        | 1.92G |
|-----------------------------------------------------------------------|
|                      System information: Data                         |
|-----------------------------------------------------------------------|
| Tier           | Capacity | Speed   | Type | Location on compute/login |
|----------------|----------|---------|------|---------------------------|
| Scratch        | 0.6PB    | 400GB/s | NVMe | /project/scratch          |
| Home/Project   | 12.5PB   | 180GB/s | HDD  | /home/users, /project/home |
| Backup         | 7.5PB    | 30GB/s  | HDD  | -                         |
|-----------------------------------------------------------------------|
|                      System information: Interconnect                 |
|-----------------------------------------------------------------------|
| Fabric: Infiniband HDR, 200Gbps, DragonFly+ topology                  |
| Links : 1x on CPU nodes, 2x on GPU, FPGA & LargeMemory nodes          |
|-----------------------------------------------------------------------|
|                      System information: Software                     |
|-----------------------------------------------------------------------|
| Production software stack: 2021.3, current default and obsolete on    |
| 14 Jan 2023                                                           |
|                                                                       |
| Production software stack: 2022.1, default from 15 Jan 2023, use it   |
| now with: `module load env/release/2022.1`                            |
|                                                                       |
|                                                                       |
| Modules system: LMod, use `module av` on nodes to discover the environment |
|-----------------------------------------------------------------------|
|                          Center information                           |
|-----------------------------------------------------------------------|
| News & Events      : luxprovide.lu                                    |
| Documentation      : docs.lxp.lu                                      |
| System status      : weather.lxp.lu                                   |
| Support            : servicedesk.lxp.lu, servicedesk@lxp.lu           |
|-----------------------------------------------------------------------|
| LinkedIn & Twitter : @luxprovide #meluxina @EuroHPC_JU                |
`-----------------------------------------------------------------------'


#######################################################################
#                  System events, in-progress and upcoming            #
#---------------------------------------------------------------------#
# * 2024-02-15 00:00 CET: New 2023.1 software stack default on compute nodes #
#     See https://docs.lxp.lu/system/whatsnew/#software-stacks for details #
#######################################################################

Last login: Tue Jan 30 16:11:10 2024 from 158.64.79.9
[u100054@login02 ~]$
```

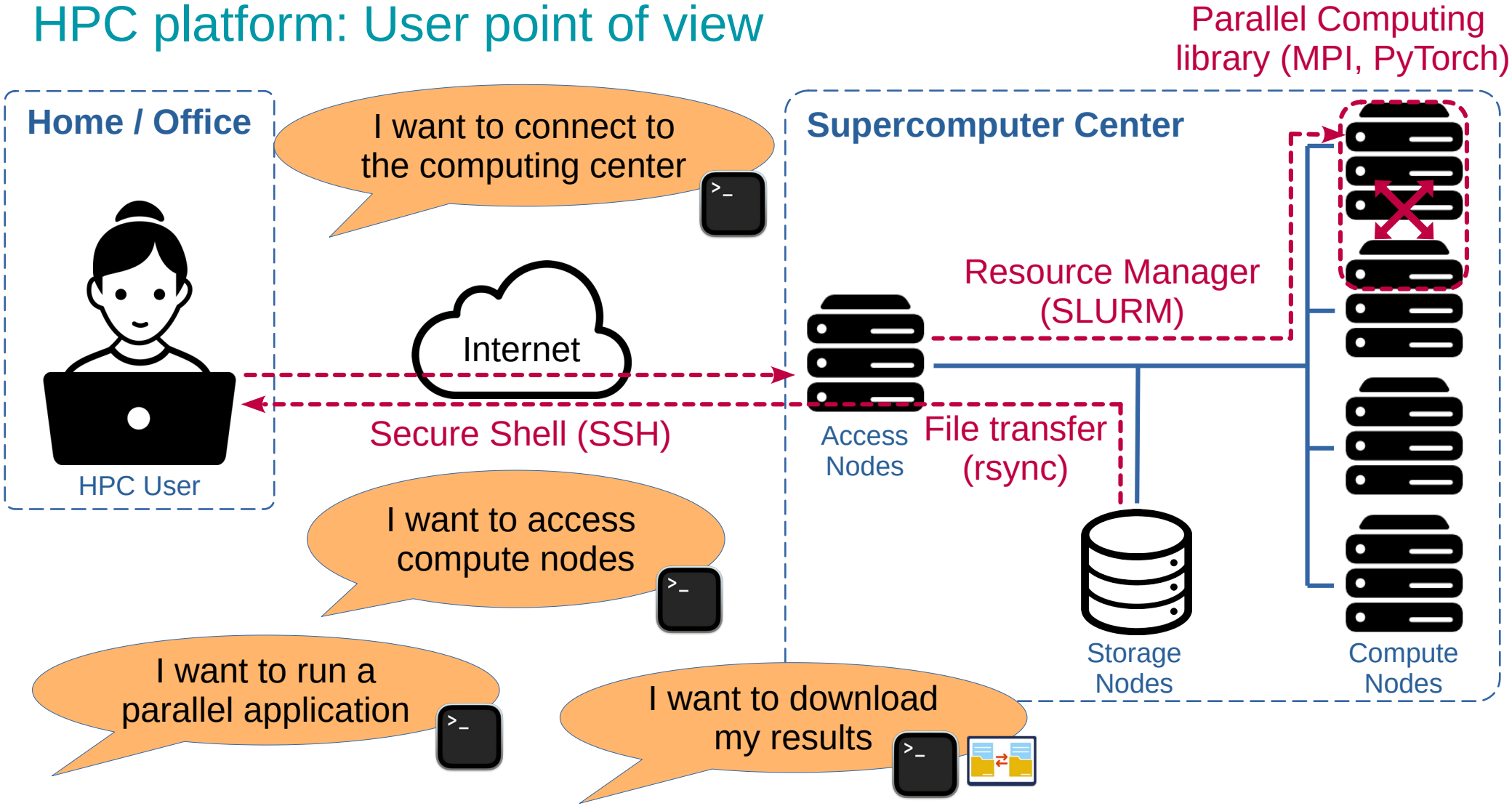
Web-portal Interface







HPC platform: User point of view



HPC usage: **Interactive** vs **Batch**

Interactive jobs

- ***Immediate*** access to compute resources
- Commands are run manually (e.g. via an interactive shell, Jupyter notebook, ...)
- Used for debugging, development, testing, or small exploratory runs
- Usually limited to small number of resources and a short time

Batch jobs (non-interactive)

- Execution is **deferred** until resources become available
- No user interaction during runtime
- Used for production workloads, long simulations, large parameter sweeps, ...
- Can use large number of nodes and long execution time
- ➔ This is the most standard usage

➢ In both cases, resource requests (or jobs) are submitted to the **resource manager**

- Traditionally, HPC do offer infinite jobs or services

slurm
workload manager

SLURM: Simple Linux Utility for Resource Management

What is SLURM?

- Open-source, highly configurable, and fault-tolerant workload manager for Linux clusters
- Widely used in HPC to manage and schedule large-scale computational jobs, including on MeluXina

Main functions

- **Resource allocation**: It allocates resources like CPUs, memory, and storage to submitted jobs based on their requirements and configured policies.
- **Job scheduling**: It queues and prioritizes jobs, ensuring fair access and efficient utilization of resources.

Main concepts of SLURM

Partitions

- Group of **nodes with similar characteristics** (CPU type, memory capacity, available software)
- Users submit jobs to specific partitions based on their resource requirements

QoS (Quality of Service)

- QoS defines **scheduling policies** for jobs within a partition, influencing priority and resource allocation
- For example, limits on job wait time and duration, pre-emption rules

Jobs

- Jobs are the **units of work** that are submitted to SLURM.
- Each job has a specific resource requirement, such as number of CPUs, memory, and disk space.
- Jobs are submitted within specific partitions and QoS levels.

Queue

- **Holding area** for submitted jobs within a partition and QoS
- Jobs are added to the queue according to arrival time and priority rules defined by the QoS
- Jobs can be PENDING, RUNNING, COMPLETED (and many more)

Job Scheduling Simple Example

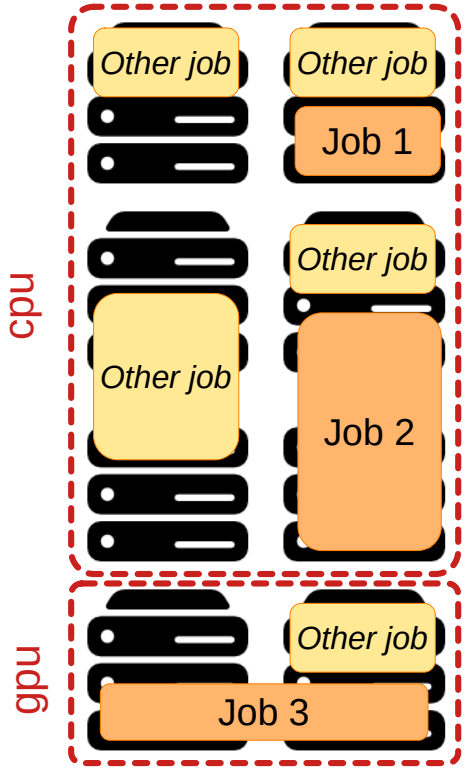
QoS:

`default`	`long`
- max 100 jobs	- max 1 job
- max 2 days	- max 6 days
- priority: normal	- priority: low

Queues:

Pending:	Job 1 PENDING **priority: 1**	Job 2 PENDING **priority: 99**	Job 3 PENDING **priority: 10**
Running:	Job 1 RUNNING	Job 3 RUNNING	Job 2 RUNNING
Completed:	Job 1 COMPLETED	Job 3 COMPLETED	Job 2 COMPLETED

Partitions



Compute Nodes

HPC User submits jobs

```
salloc -A p20XXXX -N 1 -t 2:00 -p cpu -q default
```
→ *Job 1 for 1 CPU node for 2 hours*

```
salloc -A p20XXXX -N 4 -t 6-0  -p cpu -q long
```
→ *Job 2 for 4 CPU nodes for 6 days*

```
salloc -A p20XXXX -N 2 -t 8:00 -p gpu -q default
```
→ *Job 3 for 2 GPU nodes for 8 hours*

Find and Using Software on HPC with Lmod

Software Modules on HPC

- Manage **user environment** using Environment Modules
- Search and make software available to users
- Using `PATH`, `LD_LIBRARY_PATH` and other environment variables
- Various implementations: **Lmod**, C and Tcl Environment Modules

```
[         @mel0319 ~]$ module avail GROMACS

-------- /apps/USE/easybuild/release/2025.1/modules/all --------
   GROMACS/2025.2-foss-2025a-CUDA-12.8.0
   GROMACS/2025.2-foss-2025a             (D)

  Where:
   D:  Default Module
```

`module avail`: Lists all available modules

`module list`: Shows the loaded modules

`module load <module_name>`: Loads a module to make the software accessible

`module unload <module_name>`: Removes a module from your environment

Other useful commands: `overview`, `spider`, `swap`, `help`, `purge`, `keyword`, …

➔ https://docs.lxp.lu/first-steps/module_system/

MeluXina in practice

What you need to know to use MeluXina

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MeluXina user and project accounts

User account: `u10XXXX`

It identifies a **person** (you)

- First and last name
- Email

It serves for authentication with

- Password
- One-Time Password (OTP)
- Authorized SSH keys

Datapath:

`/home/users/u10XXXX`

Project account: `p20YYYY`

It identifies a **project** (that you can be part of)

- Group of users
- Start and end time

It tracks the resources

- **Computing resources**:
  `cpu`, `gpu`, `fpga`, `largemem`
- **Storage resources**:
  `scratch` (Tier1) and `project` (Tier2)

Datapath:

`/project/home/p20YYYY`

`/project/scratch/p20YYYY`

MeluXina shared data storage

Shared storage accessible from any compute node

- Tier1: fast but small → temporary storage
- Tier2: slower but larger → permanent storage

User storage → personal and limited

`/home/users/u10XXXX` → located on Tier2

Project storage → shared with project members, size specific to each project

`/project/home/p20YYYY` → located on Tier2 ← **Preferred location for your work files**

`/project/scratch/p20YYYY` → located on Tier1

Tracking your resource usage → myquota

```
                    COMPUTE ALLOCATIONS FOR CURRENT MONTH
                    COMPUTE USAGE FROM  2026-04-01 to NOW

Project          CPU node-hours            GPU node-hours            FPGA node-hours
               Used   Max   %used        Used   Max   %used        Used   Max   %used
p20[redacted]   0                         0                         0
p20[redacted]   0     328    0.0%         0     65    0.0%         0
p20[redacted]   0                         8    130    6.2%         0
```

Compute

```
                                DATA ALLOCATIONS

 Datapath                     Data GiB                        No. Files
                        Used |   Max   | Use%          Used |    Max    | Use%
/home/users/[redacted]    90 |    100  | 90%          94566 |   100000  | 94%
/project/home/p20[redacted]  0 |   1000  | 0%             142 |  1000000  | 0%
/project/home/p20[redacted] 2428 |  20000 | 12%         850390 |  2000000  | 42%
/project/scratch/p20[redacted]  0 |  1024  | 0%               1 |  1000000  | 0%
/project/home/p20[redacted]  0 |  10240  | 0%            1467 |  1000000  | 0%
```

Data

Authentication & Access to MeluXina

SSH Key

- Fast, lightweight, and secure
- Allows password-less access
- Most common on HPC platforms
- Requires an advanced configuration

> Public SSH key to be upload via the service desk

Used for frequent and password-less operations

- **Shell access** via native terminal applications
- File transfer and synchronization

Username + Password (+ OTP)

- Beginner friendly, easier to setup
- Similar to usual online web-service
- Manual authentication process

> Username: **`u10XXXX`**
> Password: defined at service desk setup
> OTP: defined at first use

Used for interactive and graphical accesses

- **Web-portal access**
  (also includes a shell access)
- Simple file transfer only

Shell Access via SSH

Configuration

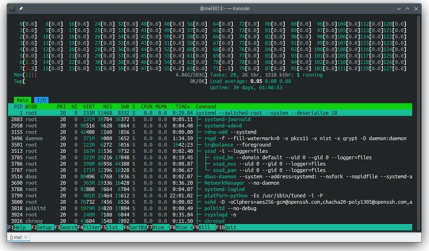
- Public SSH key to be added via service desk
- Configuration file `~/.ssh/config`

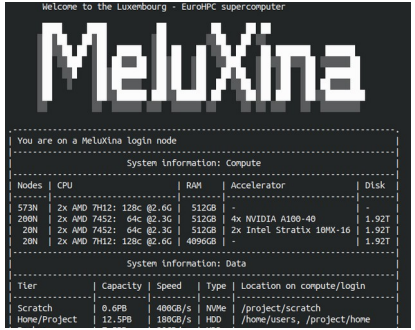
Access

```
ssh meluxina
```

Documentation

- Generating an SSH key
- Uploading your public SSH key
- Connect to MeluXina





Web-portal Access

Configuration

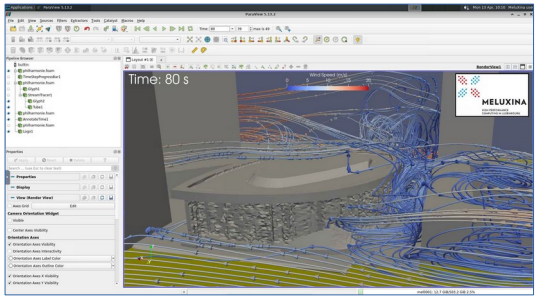
- Password setup in service desk
- One-Time Password (OTP) setup at first access

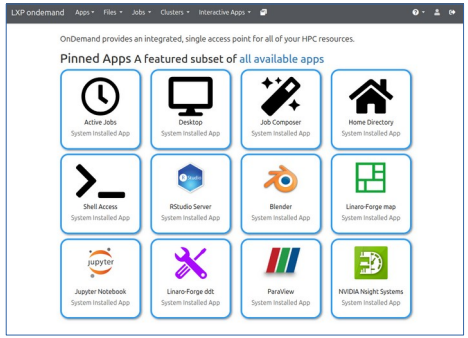
Access

https://portal.lxp.lu/

Documentation

- How to connect to Open Ondemand
- Multi-factor authentication setup





Hands-on!

https://luxprovide.github.io/SCynergy2026-GettingStartedWithMeluXina/

- Connecting the MeluXina
- HPC workflow: Urban wind simulation
- AI workflow: Deep learning with PyTorch

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